

■ Enhanced Technical Interview Notes (DSA + CS Fundamentals)

1. Data Structures & Algorithms (DSA)

- Arrays & Strings – basics + problem solving must
- Sliding Window + Two Pointers (VERY IMPORTANT)
- Stack & Queue – LRU, valid parentheses type problems
- Linked List – reverse, detect cycle
- Trees – BST, traversals (Inorder, Preorder, Postorder)
- Graphs – BFS, DFS, shortest path basics
- Dynamic Programming – start with patterns (not memorization)
- ■ Focus: Identify patterns instead of memorizing solutions

■ Must Solve Problem Patterns

- Two Sum / Subarray problems
- Binary Search variations
- Sliding Window maximum/minimum
- Tree traversal + recursion
- Graph traversal (BFS/DFS)
- DP: Fibonacci, Knapsack basics

2. Core CS Fundamentals

- OOP: Encapsulation, Inheritance, Polymorphism, Abstraction
- DBMS: Joins, Indexing, Normalization, Keys
- Operating Systems: Process vs Thread, Deadlock
- Computer Networks: HTTP vs HTTPS, TCP/IP basics

3. Problem Solving Strategy

- Understand problem → dry run first
- Start with brute force → optimize
- Write clean step-by-step logic
- Practice consistency daily (1–2 problems)
- Revise old problems weekly

4. Important Interview Questions

- What is time complexity and space complexity?
- Difference between stack and queue?
- What is hashing and where is it used?
- What is BFS vs DFS?
- What is normalization in DBMS?
- Explain deadlock conditions
- What is REST API?

■ Final Advice

- Don't memorize solutions
- Focus on thinking patterns
- Consistency > intensity
- Practice daily, even small progress matters